



Lesson 2.9 — Modeling with Quadratics

Big idea: Quadratics model anything where two effects are competing — thrust vs. gravity, length vs. width, revenue per unit vs. units sold. The vertex tells you the maximum or minimum of that trade-off.

Quick review

Projectile model: $h(t) = -16t^2 + v_0 t + h_0$ (feet) or $-4.9t^2 + v_0 t + h_0$ (meters).

Area / perimeter trade-off: with a fixed perimeter, area is maximized at a particular length-width split (the vertex of an area quadratic).

Revenue: $R = (\text{price})(\text{quantity})$. When price drops linearly raise quantity, revenue is quadratic in price (or quantity).

Workflow: (1) write the quadratic in context, (2) decide if you need vertex (max/min), zeros (when it equals zero), or specific output, (3) compute and interpret with units.

Worked example

Worked example. A baseball is hit with $h(t) = -16t^2 + 80t + 4$. Find the maximum height and when it lands.

Vertex at $t = 80/32 = 2.5$ s. Max height $= -16(2.5)^2 + 80(2.5) + 4 = 104$ ft.

Lands when $h = 0$: $16t^2 - 80t - 4 = 0 \Rightarrow t \approx 5.05$ s.

Warm-ups

1. $h(t) = -16t^2 + 64t$. When does the projectile land?
2. A rectangle has perimeter 40. If width = w , write the area $A(w)$.
3. Revenue $R(p) = p(100 - p)$. What price maximizes revenue?

Practice

1. A rocket's height is $h(t) = -16t^2 + 96t + 5$. Find the maximum height and the time it occurs.
2. A farmer has 200 ft of fencing to enclose a rectangular field along a river (no fence on the river side). What dimensions maximize area?
3. A theater sells 200 tickets at \$10 each. For every \$1 increase, 10 fewer tickets sell. Write revenue $R(x)$ where x = number of \$1 increases. Find the price that maximizes revenue.

Challenge

1. A ball is thrown from a 6 ft platform with initial velocity 32 ft/s. (a) Write $h(t)$. (b) Find max height. (c) When does it hit the ground?
2. A swimming pool company's profit (in \$1000s) on x pools per month is $P(x) = -0.5x^2 + 30x - 200$. (a) How many pools maximize profit? (b) What is that max profit? (c) For what range of x do they break even or better?



Answer key — Lesson 2.9

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $h = 0 \Rightarrow -16t(t - 4) = 0 \Rightarrow t = 0$ (start) or $t = 4$ s (lands).
2. Length = $20 - w$. $A(w) = w(20 - w) = 20w - w^2$.
3. Vertex at $p = 50$. Max revenue = $50 \cdot 50 = \$2,500$.

Practice

1. Vertex $t = 96/32 = 3$ s. $h(3) = -144 + 288 + 5 = 149$ ft.
2. If two widths w + length $L = 200 \Rightarrow L = 200 - 2w$. $A(w) = w(200 - 2w) = 200w - 2w^2$. Vertex $w = 50$ ft, $L = 100$ ft. Max area 5,000 sq ft.
3. Tickets sold: $200 - 10x$. Price: $10 + x$. $R(x) = (10 + x)(200 - 10x) = -10x^2 + 100x + 2000$. Vertex $x = 5 \rightarrow$ price \$15. Max revenue = \$2,250 (at 150 tickets).

Challenge

1. (a) $h(t) = -16t^2 + 32t + 6$. (b) $t = 1$ s, max = $-16 + 32 + 6 = 22$ ft. (c) $-16t^2 + 32t + 6 = 0 \Rightarrow t = \frac{-32 \pm \sqrt{(1024 + 384)}}{-32} = \frac{32 \pm \sqrt{1408}}{32} \approx \frac{32 \pm 37.52}{32}$. Positive: $t \approx 2.17$ s.
2. (a) Vertex $x = 30$ pools. (b) $P(30) = -450 + 900 - 200 = \$250,000$. (c) $P \geq 0 \Rightarrow 0.5x^2 - 30x + 200 \leq 0$; zeros at $x = \frac{30 \pm \sqrt{(900 - 400)}}{1} = 30 \pm \sqrt{500} \approx 30 \pm 22.36$. So roughly $8 \leq x \leq 52$ pools (must be whole pools).